Supplementary material

Contents

1 Case study results

Table S1. Results of reconstruction for each fold on the Carnivora dataset

Fold	AUC	Threshold	Nr. Held-out	Pred. held-out ones	Total pred. ones
1	0.99	0.0015	178	0.49	3378
2	0.98	0.0016	178	0.38	2849
3	0.99	0.0022	178	0.36	2583
4	0.99	0.002	178	0.46	3160
5	0.99	0.0016	178	0.47	3224
6	0.99	0.0024	179	0.31	2741
7	0.99	0.0019	178	0.34	2589
8	0.99	0.0019	178	0.48	3262
9	0.99	0.0015	179	0.39	2784
10	0.99	0.001	179	0.45	3388
Average	**0.99**	**0.0018**		**0.41**	**2995.8**

Table S2. Literature search for most likely interactions

Host	Parasite	Prob	SD	Evidence	Ref
Neovison vison	Mesocestoides lineatus	0.19	0.27	Confirmed	Zschille et al. (2004)
Lynx rufus	Carnivore protoparvovirus 1	0.12	0.31	Confirmed	Allison et al. (2013)
Procyon lotor	Yersinia pestis	0.08	0.08	Confirmed	Clover et al. (1989)
Meles meles	Eucoleus aerophilus	0.06	0.06	Confirmed	Byrne et al. (2020)
Lynx rufus	Felid alphaherpesvirus 1	0.06	0.04	Plausible. Artificial innoculation resulted in asymptomatic infection.	Eberle et al. (1991)
Procyon lotor	Ctenocephalides felis	0.04	0.07	Confirmed	Sharifdini et al. (2021)
Nyctereutes procyonoides	Capillaria aerophila	0.03	0.03	Confirmed for parasite synonym name *Eucoleus aerophilus*	Laurimaa et al. (2016)
Lutra lutra	Molineus patens	0.03	0.03	Confirmed	Takeuchi-Storm et al. (2021)
Nyctereutes procyonoides	Toxoplasma gondii	0.03	0.02	Confirmed	Osten-Sacken et al. (2024)
Neovison vison	Macracanthorhynchus catulinus	0.03	0.02	No evidence found. Infection of other members of Mustela genus.	Shimalov and Shimalov (2002)

Table S3. Mean phylogenetic distance for grouping levels found for hosts during Hierarchical Stochastic Block Model inference for fold 6 on the Carnivora dataset. Results shown for groups with at least 10 hosts.

Level	Nr hosts	Obs.	Null mean	Null sd	p	p<0.05
Level 1 (nr groups: 12)						
	32	102.419	101.197	2.166	0.669	
	19	61.710	101.346	3.175	0.001	*
	22	103.088	101.360	2.818	0.704	
	10	82.213	101.422	5.928	0.014	*
	15	88.617	101.362	3.988	0.011	*
Level 2 (nr groups: 4)						
	69	99.232	101.318	1.058	0.040	*
	27	61.427	101.371	2.351	0.001	*
	39	83.234	101.240	1.824	0.001	*
Level 3 (nr groups: 2)						
	135	100.806	101.258	0.194	0.019	*
	135	100.806	101.273	0.188	0.012	*

Table S4. Mean nearest taxon distance for grouping levels found for hosts during Hierarchical Stochastic Block Model inference for fold 6 on the Carnivora dataset. Results shown for groups with at least 10 hosts.

Level	Nr hosts	Obs	Null mean	Null sd	p	p<0.05
Level 1 (nr groups: 12)						
	32	26.387	25.356	3.119	0.623	
	19	23.968	31.671	5.382	0.076	
	22	28.264	29.647	4.274	0.372	
	10	22.180	44.657	10.077	0.014	*
	15	25.960	35.553	6.664	0.066	
Level 2 (nr groups: 4)						
	69	19.928	18.884	1.347	0.787	
	27	22.459	27.296	3.615	0.094	
	39	15.087	23.578	2.579	0.002	*
Level 3 (nr groups: 2)						
	135	15.216	15.358	0.262	0.288	
	135	15.216	15.354	0.262	0.279	

Table S5. Mean phylogenetic distance of inferred groupings compared to null model. G1, G2, G3, G4 are the grouping levels. In each level the numbers refer to how many inferred grouping were found to be significant (p $<$ 0.05) compared to a null model, over all groupings with more than 10 taxa. So 2/4 means that two inferred groupings were found to be significantly clustered on a total of 4 inferred groupings with more than 10 taxa.

Fold	**G1**	**G2**	**G3**	**G4**
1	3/6	2/3	1/1	1/1
2	2/5	2/3	0/1	NA
3	2/4	2/4	0/1	0/1
4	2/4	1/3	0/1	NA
5	2/4	4/4	1/1	NA
6	3/5	2/3	1/1	1/1
7	4/6	2/3	0/1	NA
8	2/5	3/3	0/1	NA
9	2/4	2/3	0/1	NA
10	3/5	4/4	0/1	NA
Average	**0.52**	**0.72**	**0.3**	**0.67**

Table S6. Mean nearest taxon distance of inferred groupings compared to null model. G1, G2, G3, G4 are the grouping levels. In each level the numbers refer to how many inferred grouping were found to be significant (p $<$ 0.05) compared to a null model, over all groupings with more than 10 taxa. So 2/4 means that two inferred groupings were found to be significantly clustered on a total of 4 inferred groupings with more than 10 taxa.

Fold	**G1**	**G2**	**G3**	**G4**
1	0/6	1/3	0/1	0/1
2	0/5	0/3	0/1	NA
3	0/4	1/4	0/1	0/1
4	0/4	0/3	1/1	NA
5	1/4	0/4	0/1	NA
6	1/5	1/3	0/1	0/1
7	2/6	1/3	0/1	NA
8	1/5	2/3	1/1	NA
9	0/4	0/3	0/1	NA
10	0/5	1/4	0/1	NA
Average	**0.1**	**0.22**	**0.2**	**0**

2 Mathematical framework for network reconstruction

We provide a brief description of the nonparametric Bayesian framework for network reconstruction implemented in the `MeasuredBlockModel` class of the *graph-tool* (http://graph-tool.skewed.de/) Python module, which serves as the inference engine used by the *sabinaHSBM* R package. A complete description of the framework can be found in the works of Peixoto (2017, 2018).

For the network reconstruction task, the model assumes that the observed network $\boldsymbol{D}$ represents the best available information about the true underlying network $\boldsymbol{A}$. Each entry $D_{ij} = 1$ indicates that an interaction was observed between node pairs (i, j), whereas $D_{ij} = 0$ denotes either an unobserved interaction or a link temporarily removed as part of a cross-validation procedure. Due to sampling limitations, observational biases and recording errors, $\boldsymbol{D}$ is treated as a noisy observation of $\boldsymbol{A}$ potentially containing both false positives (spurious links) and false negatives (missing links).

The objective is to reconstruct the true and unobserved binary adjacency matrix $\boldsymbol{A}$ (where $A_{ij} \in \{0, 1\}$) from the set of noisy observations in $\boldsymbol{D}$. Let n_{ij} denote the total number of independent observation trials for the pair (i, j), and x_{ij} denote the number of times an interaction was recorded. For common binary incidence matrices without information on measurement trials, this can be simply reduced to $n_{ij} = 1$ and $x_{ij} = D_{ij}$ (Peixoto, 2018). The inference of the true network is performed jointly with the inference of its mesoscale structure, represented by the parameters $\boldsymbol{\theta}$ of a Hierarchical degree-corrected Stochastic Block Model (hereafter denoted as HSBM). The joint posterior distribution of the network, HSBM parameters, and measurement model hyperparameters is given by:

$$P(\boldsymbol{A}, \boldsymbol{\theta}, \boldsymbol{\Omega}|\boldsymbol{x}, \boldsymbol{n}) = \frac{P(\boldsymbol{x}|\boldsymbol{n}, \boldsymbol{A}, \boldsymbol{\Omega})P(\boldsymbol{A}|\boldsymbol{\theta})P(\boldsymbol{\theta})P(\boldsymbol{\Omega})}{P(\boldsymbol{x}|\boldsymbol{n})} \tag{1}$$

Here $\boldsymbol{\theta}$ denotes the latent parameters of the HSBM, while $\boldsymbol{\Omega}$ encapsulates the hyperparameters of the measurement model. This formulation factorizes the problem into four components:

1. Measurement model $P(\boldsymbol{x}|\boldsymbol{n}, \boldsymbol{A}, \boldsymbol{\Omega})$: the likelihood of generating the observed data given the hypothesized network and measurement error parameters. This corresponds to a beta-binomial measurement process that models false positive and false negative error rates.

2. Structural prior $P(\boldsymbol{A}|\boldsymbol{\theta})$: the probability of the true network $\boldsymbol{A}$ existing under the HSBM generative process, conditioned on the inferred hierarchical block structure.

3. Model prior $P(\boldsymbol{\theta})$: the prior probability assigned to the hierarchical block structure. Because the microcanonical ensemble (explained in Section 2.2.2) deterministically fixes the structural parameters given the network and the block partition, this term is entirely governed by the nonparametric prior over the block partitions. It acts as a complexity penalty, penalizing overly complex representations to prevent overfitting.

4. Measurement prior $P(\boldsymbol{\Omega})$: the prior distribution over the measurement error parameters of the false positive and false negative rates.

In the absence of a structural prior, and assuming uniform error rates across all node pairs, the model can only estimate global average probabilities for missing or spurious links. By incorporating the HSBM prior, the inferred parameters $\boldsymbol{\theta}$ define block partitions that allow the model to borrow statistical strength across structurally similar nodes. Consequently, individual link probabilities are no longer estimated independently, but instead become informed by the mesoscale structure of the network.

A key advantage of this framework is that the inference of the true network $\boldsymbol{A}$ and the inference of the community structure are treated as a single joint Bayesian inference problem, rather than as two sequential procedures. This allows uncertainty in the network reconstruction and uncertainty in the mesoscale structure to inform one another during inference.

Because the microcanonical HSBM framework performs Bayesian model selection through minimum description length minimization (see 2.3), it naturally penalizes overly complex models. Additional groups or hierarchical levels are introduced only when the improvement in explanatory power outweighs the increase in model complexity. As a result, the model is particularly well suited for distinguishing genuine mesoscale structure from random fluctuations, which is especially important when analysing sparse and noisy ecological data.

Ultimately, the use of a structured generative prior provides an essential form of regularization. Whenever the underlying network exhibits non-random patterns, the HSBM offers opportunities to mathematically compress the network representation and reduce the overall description length. Beyond aiding in accurate link prediction, the inferred community structure itself provides biologically meaningful information for downstream ecological interpretation and analysis (Peixoto, 2025).

2.1 The measurement model

Let $x_{ij} \in \{0, 1, ..., n_{ij}\}$ be the number of times an interaction was observed across n_{ij} independent measurements (trials).

In *sabinaHSBM* we employ the `MeasuredBlockModel` class in *graph-tool* which assumes a uniform observation error model characterized by two global parameters:

- p: the false negative rate, i.e. the probability of missing a true interaction in a single trial.
- q: the false positive rate, i.e. the probability of recording a spurious interaction in a single trial.

Notice that p and q are constant across all node pairs (i, j).

Given a hypothesized true network $\boldsymbol{A}$, the likelihood of recording exactly $\boldsymbol{x}$ interactions over $\boldsymbol{n}$ trials follows a product of binomial distributions:

$$P(\boldsymbol{x}|\boldsymbol{n}, \boldsymbol{A}, p, q) = \prod_{i<j} \binom{n_{ij}}{x_{ij}} \left[(1-p)^{x_{ij}} p^{n_{ij}-x_{ij}}\right]^{A_{ij}} \left[q^{x_{ij}}(1-q)^{n_{ij}-x_{ij}}\right]^{1-A_{ij}} \tag{2}$$

Because error rates are rarely known, fixing p and q as point estimates risks overfitting. To prevent this, the global error rates are treated as nuisance parameters and integrated out over their probability distributions:

$$P(\boldsymbol{x}|\boldsymbol{n}, \boldsymbol{A}, \alpha, \beta, \mu, \nu) = \int_0^1 \int_0^1 P(\boldsymbol{x}|\boldsymbol{n}, \boldsymbol{A}, p, q) P(p|\alpha, \beta) P(q|\mu, \nu) \, \mathrm{d}p \, \mathrm{d}q \tag{3}$$

We assign conjugate Beta prior distributions to the error rates $p \sim \mathrm{Beta}(\alpha, \beta)$ and $q \sim \mathrm{Beta}(\mu, \nu)$, defined as:

$$\begin{aligned} P(p|\alpha, \beta) &= \frac{p^{\alpha-1}(1-p)^{\beta-1}}{B(\alpha, \beta)} \\ P(q|\mu, \nu) &= \frac{q^{\mu-1}(1-q)^{\nu-1}}{B(\mu, \nu)} \end{aligned} \tag{4}$$

where $B(\cdot, \cdot)$ denotes the Euler beta function.

In the *sabinaHSBM* package we adopt the noninformative case where the hyperparameters are set to $\alpha = \beta = \mu = \nu = 1$. This corresponds to uniform $\mathrm{Beta}(1, 1)$ priors that express complete ignorance about measurement noise. Under this choice, the marginal likelihood is fully governed by four global sums that characterize how the data maps onto the proposed network $\boldsymbol{A}$:

$$\begin{aligned} \mathcal{M} &= \sum_{i<j} n_{ij} \quad \text{(Total number of trials)} \\ \mathcal{X} &= \sum_{i<j} x_{ij} \quad \text{(Total number of recorded interactions)} \\ \mathcal{E} &= \sum_{i<j} n_{ij} A_{ij} \quad \text{(Total number of trials on true edges)} \\ \mathcal{T} &= \sum_{i<j} x_{ij} A_{ij} \quad \text{(Total number of recorded true edges)} \end{aligned}$$

By integrating out the hyperparameters under the noninformative prior, the likelihood simplifies into a closed-form combinatorial expression:

$$P(\boldsymbol{x}|\boldsymbol{n}, \boldsymbol{A}) = \left[\prod_{i<j} \binom{n_{ij}}{x_{ij}}\right] \left[\frac{1}{\mathcal{E}+1} \binom{\mathcal{E}}{\mathcal{T}}^{-1}\right] \left[\frac{1}{(\mathcal{M}-\mathcal{E})+1} \binom{\mathcal{M}-\mathcal{E}}{\mathcal{X}-\mathcal{T}}^{-1}\right] \tag{5}$$

2.1.1 Reconstruction of single trial networks (binary networks)

In many empirical networks, repeated observation trials ($\boldsymbol{n}$) and the number of recorded interactions ($\boldsymbol{x}$) are unavailable. Instead, most networks are presented as binary adjacency matrices $\boldsymbol{D}$ where an interaction was either observed $D_{ij} = 1$ or not $D_{ij} = 0$, with no information on measurement effort. Following Peixoto (2018), the *sabinaHSBM* package treats this scenario as a special case of the measurement model by assuming a single trial for all pairs. Under this setting $n_{ij} = 1$ for all node pairs, and the number of recorded interactions corresponds to the entries of the observed adjacency matrix $x_{ij} = D_{ij} \in \{0, 1\}$. Because $n_{ij} = 1$, the binomial coefficient $\binom{n_{ij}}{x_{ij}}$ strictly equals 1 and drops out of the equation (2). The binomial measurement model simplifies into a bernoulli error model:

$$P(\boldsymbol{x}|\boldsymbol{n},\boldsymbol{A},p,q)=\prod_{i<j}[(1-p)^{x_{ij}}p^{1-x_{ij}}]^{A_{ij}}[q^{x_{ij}}(1-q)^{1-x_{ij}}]^{1-A_{ij}} \tag{6}$$

In this scenario, the four global sums described previously can be mapped onto the standard components of a binary confusion matrix comparing the observed network $\boldsymbol{D}$ to the hypothesized true network $\boldsymbol{A}$:

- True Positives ($TP=\mathcal{T}$): Edges present in both $\boldsymbol{D}$ and $\boldsymbol{A}$.
- False Negatives ($FN=\mathcal{E}-\mathcal{T}$): Edges absent in $\boldsymbol{D}$ but present in $\boldsymbol{A}$.
- False Positives ($FP=\mathcal{X}-\mathcal{T}$): Edges present in $\boldsymbol{D}$ but absent in $\boldsymbol{A}$.
- True Negatives ($TN=\mathcal{M}-\mathcal{E}-FP$): Edges absent in both $\boldsymbol{D}$ and $\boldsymbol{A}$.

After integrating out the global error rates p and q using the non-informative uniform priors ($\text{Beta}(1,1)$), the marginal likelihood of the binary observation network simplifies entirely to:

$$P(\boldsymbol{x}|\boldsymbol{n},\boldsymbol{A})=\left[\frac{1}{TP+FN+1}\binom{TP+FN}{TP}^{-1}\right]\left[\frac{1}{FP+TN+1}\binom{FP+TN}{FP}^{-1}\right] \tag{7}$$

In the single-trial specification, the measurement model alone suffers from perfect symmetry, i.e. all unobserved edges ($D_{ij}=0$) are statistically indistinguishable, meaning the data assigns them identical probabilities of being false negatives. Consequently, link prediction is impossible using the measurement model in isolation. However, by coupling the measurement model with the Hierarchical Stochastic Block Model (HSBM), the algorithm breaks this symmetry by borrowing statistical strength from the mesoscale topology. The HSBM pools node pairs through their inferred group memberships. For example, if the block structure implies a high structural probability of interaction between two groups, a missing edge between nodes in those groups becomes more likely as a false negative, overriding the weak evidence of the single zero.

Because the single-trial binary matrix represents the vast majority of available network datasets, it is the default configuration in the *sabinaHSBM* package. Users interested in modelling varying measurement efforts can provide custom values of n_{ij} and x_{ij} via a dataframe passed to the `custom_nx` argument of the `hsbm.input` function, enabling extensions beyond the single-trial assumption (more details in Section 4).

2.2 The structural prior

The structural prior $P(\boldsymbol{A}|\boldsymbol{\theta})$ evaluates the probability that the hypothesized true network $\boldsymbol{A}$ is consistent with the generative rules of the Stochastic Block Model (SBM). Following Peixoto (2017) framework, this is formalized using the microcanonical ensemble, which constrains the model to generate only graphs that perfectly satisfy the supplied parameters, assigning a uniform probability to all valid configurations.

In the standard SBM (Holland et al., 1983), the generative parameters are $\boldsymbol{\theta}=\{\boldsymbol{b},\boldsymbol{e}\}$. Here, $\boldsymbol{b}$ is the vector mapping the N nodes into B blocks, where $b_i\in\{1,...,B\}$ specifies the group membership of node i. The parameter $\boldsymbol{e}$ is a $B\times B$ matrix where e_{rs} is the number of edges between block r and block s. This can be defined as:

$$e_{rs}=\sum_{ij}A_{ij}\delta_{b_i,r}\delta_{b_j,s}$$

(or twice that number for internal block edges $r=s$). The δ is the Kronecker delta, and acts as a logical filter ensuring that only edges connecting a node in block r to a node in block s contribute to the sum.

A critical limitation of the standard SBM is that it assumes all nodes within a group have roughly the same degree. In most networks this is rarely true. For instance, when applied to a highly nested network plant-pollinator network, the model tends to partition nodes by their degree, artificially separating generalist species from specialist species.

To resolve this, the degree-corrected SBM extends the parameter set to $\boldsymbol{\theta}=\{\boldsymbol{b},\boldsymbol{e},\boldsymbol{k}\}$, where $\boldsymbol{k}$ is the exact degree sequence of the true network $\boldsymbol{A}$. Conditioning the structural prior on $\boldsymbol{k}$ preserves heterogeneous degree distributions. In the degree-corrected SBM, the expected number of interactions between node i and j scales with their degrees k_i and k_j, allowing the model to discover modular structure without erasing core-periphery (e.g. nested) degree distributions.

2.2.1 Canonical form of the structural prior

The model assumes the true network is a multigraph where the number of interactions A_{ij} between any two nodes is drawn from a Poisson distribution. While most empirical interaction networks are recorded as binary simple graphs ($A_{ij} \in \{0,1\}$), the Poisson multigraph is asymptotically identical to a Bernoulli simple graph in the sparse limit (Peixoto, 2017). Furthermore, once we group nodes into blocks, the connections between those blocks inherently form a multigraph, as the entries in the block matrix correspond to the aggregate number of edges between groups. Therefore, to correctly model the hierarchical structure of the network, the latent parameters must operate in a multigraph space.

The parameters of the model in the canonical form are $\{\boldsymbol{b}, \boldsymbol{\omega}, \boldsymbol{\gamma}\}$, where:

- $\boldsymbol{b}$ is the vector of block assignments (b_i is the block of node i).
- $\boldsymbol{\omega}$ is a $B \times B$ matrix where ω_{rs} represents the expected number of edges between block r and block s.
- $\boldsymbol{\gamma}$ is a vector of node-specific degree-correction parameters, representing the propensity of node i to form edges. This is normalized such that for all blocks r, $\sum_i \gamma_i \delta_{b_i,r} = 1$, where δ is again the Kronecker delta.

Under this formulation, the number of edges A_{ij} between any two nodes i and j is drawn from independent Poisson distributions. The expected number of edges (the Poisson rate parameter λ_{ij}) is the product of their expected number of edges between blocks and their individual degree propensities:

$$\lambda_{ij} = \gamma_i \gamma_j \omega_{b_i b_j} \tag{8}$$

Because most networks are inherently sparse, $\gamma_i \gamma_j \omega_{b_i b_j}$ is typically very small. Consequently, the probability of generating a multiedge ($A_{ij} \geq 2$) vanishes, allowing this Poisson multigraph ensemble to act as an accurate approximation of a simple binary graph.

To understand how the expected interaction rate λ_{ij} behaves, imagine a plant species (i) and a pollinator species (j). The parameter $\omega_{b_i b_j}$ establishes the expected block-level edge counts—the expected number of interactions between the plant's block b_i and the pollinator's block b_j. Suppose this number is 200. To determine the rate for these specific species, this value is scaled by their individual interaction frequencies ($\boldsymbol{\gamma}$). Because $\boldsymbol{\gamma}$ is normalized to sum to 1 within a block, it acts as a fraction. The parameter γ_i represents the interaction propensity of the plant i. Suppose this plant is a generalist accounting for half of all interactions in its block, so $\gamma_i = 0.5$. The parameter γ_j represents the interaction propensity of the pollinator j. Suppose it is an infrequent visitor, accounting for 10% of its block interactions, so $\gamma_j = 0.1$. The expected number of interactions between these two specific species is the product of their individual interaction frequencies and their block-level edge counts: $0.5 \times 0.1 \times 200 = 10$ expected interactions.

With this, we can formulate the full canonical structural likelihood for the true network, given the $\{\boldsymbol{b}, \boldsymbol{\omega}, \boldsymbol{\gamma}\}$ parameters, using the product of Poisson probabilities over all pairs:

$$P(\boldsymbol{A}|\boldsymbol{b}, \boldsymbol{\omega}, \boldsymbol{\gamma}) = \prod_{i<j} \frac{e^{-\lambda_{ij}} \lambda_{ij}^{A_{ij}}}{A_{ij}!} \times \prod_{i} \frac{e^{-\lambda_{ii}/2} (\lambda_{ii}/2)^{A_{ii}/2}}{(A_{ii}/2)!} \tag{9}$$

Notice that the equation is split into two factors. The first governs inter-node interactions (off-diagonal). The second explicitly models intra-node interactions (the diagonal). This allows the formula to natively handle self-loops, such as intra-specific competition or cannibalism, by halving the rate and edge count to avoid counting the same undirected physical interaction twice.

2.2.2 Microcanonical form of the structural prior

The canonical and microcanonical forms of the SBM provide two equivalent representations of the same underlying model, differing only in how uncertainty is expressed. In the canonical formulation, edges are generated independently according to Poisson distributions parameterized by continuous variables ($\boldsymbol{\omega}, \boldsymbol{\gamma}$), leading to a probabilistic description in terms of expected edge counts. In contrast, the microcanonical formulation conditions on the exact realizations of these quantities, namely the exact block-to-block edge counts $\boldsymbol{e}$ and the exact degree sequence $\boldsymbol{k}$, and assigns equal probability to all graphs consistent with these constraints.

Formally, the microcanonical ensemble can be obtained by integrating out the canonical parameters under non-informative priors. This marginalization replaces the continuous variability of the Poisson rates with fixed combinatorial constraints, yielding a uniform distribution over a restricted set of admissible graphs. As a result, both formulations encode the same structural assumptions, but the microcanonical model does so in terms of discrete combinatorics, avoiding overfitting and enabling efficient inference.

A useful representation of the generative process of the model is in terms of stubs (or half-edges): each node i is assigned exactly k_i stubs, representing the endpoints of future edges. The network is formed by randomly pairing these stubs, subject to the constraint that exactly e_{rs} edges connect nodes in block r to nodes in block s Thus, the randomness operates within a restricted ensemble bounded by the block-level interaction patterns and the individual node interaction propensities. In the context of ecological networks, these stubs can be interpreted as the interaction capacity of each species. Interactions arise from randomly pairing these capacities within the constraints imposed by species interaction propensity (where generalist species have more stubs than specialist species) and group-level interaction patterns.

Within this framework the resulting microcanonical likelihood of observing a specific multigraph $\boldsymbol{A}$ is

$$P(\boldsymbol{A}|\boldsymbol{b},\boldsymbol{e},\boldsymbol{k}) = \frac{\prod_{r<s} e_{rs}! \prod_r e_{rr}!! \prod_i k_i!}{\prod_{i<j} A_{ij}! \prod_i A_{ii}!! \prod_r e_r!} \tag{10}$$

where $e_r = \sum_s e_{rs}$ is the total number of stubs in block r, and $x!! = x(x-2)(x-4)\cdots$ denotes the double factorial.

This expression assigns equal probability to all valid stub matchings. It arises from counting the number of stub permutations consistent with $\boldsymbol{A}$, normalized by the total possibility space of block-level matchings. The numerator calculates the total ways to arrange the individual stubs ($k_i!$) and the edges between the blocks ($e_{rs}!$ and $e_{rr}!!$). The denominator normalizes by the total permutations of stubs within each block ($e_r!$) while correcting for the mathematical indistinguishability of multigraphs and self-loops ($A_{ij}!$ and $A_{ii}!!$).

The generative probability of the network given only the partition, $P(\boldsymbol{A}|\boldsymbol{b})$, is then defined by the joint probability of the network $\boldsymbol{A}$ and its topological constraints: the matrix of block-to-block edge counts $\boldsymbol{e}$ and the specific node degree sequence $\boldsymbol{k}$. Because the structure of $\boldsymbol{A}$ strictly dictates $\boldsymbol{e}$ and $\boldsymbol{k}$, we can factorize the probability into three distinct generative steps:

$$P(\boldsymbol{A}|\boldsymbol{b}) = \underbrace{P(\boldsymbol{A}|\boldsymbol{b},\boldsymbol{e},\boldsymbol{k})}_{\text{Link Formation}} \times \underbrace{P(\boldsymbol{k}|\boldsymbol{e},\boldsymbol{b})}_{\text{Degrees}} \times \underbrace{P(\boldsymbol{e}|\boldsymbol{b})}_{\text{Block Topology}} \tag{11}$$

Each factor can be described as:

- Link Formation $P(\boldsymbol{A}|\boldsymbol{b},\boldsymbol{e},\boldsymbol{k})$: Governs the physical wiring of the network. This is the microcanonical likelihood defined in equation (10).
- Degrees $P(\boldsymbol{k}|\boldsymbol{e},\boldsymbol{b})$: Governs the assignment of stubs to individual nodes. This assigns a uniform probability to generating a specific degree sequence $\boldsymbol{k}$, assuming all valid degree sequences that satisfy the block edge constraints $\boldsymbol{e}$ are equally likely. Detailed in Section 2.2.3.
- Block Topology $P(\boldsymbol{e}|\boldsymbol{b})$: Governs the mesoscale organization of interactions between blocks. Because this forms the foundation of the Hierarchical model, the exact derivation of $P(\boldsymbol{e}|\boldsymbol{b})$ is done using nested multigraphs, which is detailed in the Section 2.2.4.

2.2.3 Prior over the degree sequence

In the degree-corrected SBM, the structural prior requires us to evaluate the probability of the specific degree sequence $\boldsymbol{k}$. Because the microcanonical ensemble fixes the block-to-block edge counts ($\boldsymbol{e}$) and the block partitions ($\boldsymbol{b}$), we must calculate the conditional probability $P(\boldsymbol{k}|\boldsymbol{e},\boldsymbol{b})$. Since empirical networks frequently exhibit heterogeneous and skewed degree distributions, a flat prior would undesirably penalize this variability. To address this issue, the non-parametric hyperprior developed by Peixoto (2017) infers the shape of the degree distribution in two steps. First, it determines the frequency of the degrees, and second, it assigns these degrees to specific nodes. Let η_k^r denote the number of nodes in block r that possess exactly degree k. The probability of observing a specific degree sequence $\boldsymbol{k}$ is given exactly by:

$$P(\boldsymbol{k}|\boldsymbol{e},\boldsymbol{b}) = \prod_{r=1}^{B} \frac{\prod_k \eta_k^r!}{n_r!} \mathcal{P}(e_r, n_r)^{-1} \tag{12}$$

where n_r denotes the number of nodes in block r and e_r is the total degree of nodes in that block r.

The factor $\frac{\prod_k \eta_k^r!}{n_r!}$ is the inverse of the multinomial coefficient, representing the probability of the specific assignment of those degrees among the n_r distinguishable nodes within the block r.

The factor $\mathcal{P}(e_r, n_r)^{-1}$ represents the uniform probability of choosing the specific degree distribution out of all valid configurations. Here, $\mathcal{P}(e_r, n_r)$ is commonly known as the number of restricted integer partitions of e_r into at most n_r parts. It counts the number of different degree combinations whose total degree sums to e_r and involves no more than n_r nonzero entries.

2.2.4 Hierarchical sbm

The non-hierarchical degree-corrected SBM suffers from a resolution limit, whereby small communities become indistinguishable in large networks. In particular, if N is the number of nodes, the maximum number of statistically detectable groups under a noninformative prior scales approximately as $\mathcal{O}(\sqrt{N})$, which becomes highly restrictive as the network size increases.

The hierarchical SBM (HSBM) resolves this limitation by recursively modeling the block structure itself. The key observation is that the block-to-block edge count matrix $\boldsymbol{e}$ can be interpreted as the adjacency matrix of a smaller multigraph. This induced network can then be modeled by another SBM, yielding a nested multi-scale representation. As a consequence, the maximum number of detectable groups increases to $\mathcal{O}(N/\log N)$, allowing the model to resolve fine-scale structure even in large networks.

Let the observed network be partitioned into B_1 groups by the assignment vector $\boldsymbol{b}_1$. The corresponding block-to-block edge counts define a multigraph $\boldsymbol{e}_1$. At Level 1 ($l = 1$), these B_1 groups become the nodes of the multigraph, which are then partitioned into B_2 super-groups via $\boldsymbol{b}_2$, generating a coarser multigraph $\boldsymbol{e}_2$. This process is repeated recursively for L levels until a single group remains ($B_L = 1$). Now let $\{\boldsymbol{e}_l\}$ represent the full set of edge count matrices across all levels, and $\{\boldsymbol{b}_l\}$ represent the full set of partitions. The joint hierarchical prior becomes a chain of probabilities:

$$P(\{\boldsymbol{e}_l\} \mid \{\boldsymbol{b}_l\}) = \prod_{l=1}^{L} P(\boldsymbol{e}_l \mid \boldsymbol{e}_{l+1}, \boldsymbol{b}_{l+1}), \tag{13}$$

where $B_L = 1$ ensuring that the hierarchical recursion terminates when the entire network is aggregated into a single group. At the top level, the boundary condition dictates that $\boldsymbol{e}_{L+1}$ is simply the total number of edges E.

The probability $P(\boldsymbol{e}_l|\boldsymbol{e}_{l+1}, \boldsymbol{b}_{l+1})$, quantifies the number of ways in which the edges at level l can be arranged so that they aggregate consistently into the coarser structure $\boldsymbol{e}_{l+1}$. Because $\boldsymbol{e}_l$ represents a multigraph, we calculate the possibility space using combinations with replacement.

$$P(\boldsymbol{e}_l|\boldsymbol{e}_{l+1}, \boldsymbol{b}_{l+1}) = \prod_{r<s} \binom{n_r^l n_s^l + e_{rs}^{l+1} - 1}{e_{rs}^{l+1}}^{-1} \prod_{r} \binom{\frac{n_r^l(n_r^l+1)}{2} + \frac{e_{rr}^{l+1}}{2} - 1}{\frac{e_{rr}^{l+1}}{2}}^{-1} \tag{14}$$

where n_r^l is the number of groups inside block r at level l. The first term accounts for edges between distinct blocks (off-diagonal), while the second term accounts for edges within the same block (diagonal).

In this framework, the probability of individual edges is determined at the lowest level, while higher levels do not directly interact with nodes. Instead, the higher levels define a structural prior over admissible block structures, constraining the possible configurations at lower levels. Importantly, the hierarchy is inferred jointly across all levels, so it is not constructed sequentially. During sampling via MCMC, any proposed modification to the partition at a given level (e.g. node changing a group, or merging of two groups) induces changes in all higher levels, and the full hierarchical likelihood is updated accordingly. Proposals are accepted or rejected based on their effect on the overall description length, balancing model complexity against goodness-of-fit.

2.2.5 Nonparametric prior over partitions

In the full nonparametric Bayesian formulation of the hierarchical stochastic block model (Peixoto, 2017), the partition variables themselves are treated as random quantities. This requires specifying a prior distribution over partitions.

At each level l, the partition $\boldsymbol{b}_l$ assigns B_{l-1} nodes to B_l groups. Assuming all partitions with fixed group sizes are equally likely, and adopting a uniform prior over the number of groups, the nonparametric prior is given by

$$P(\boldsymbol{b}_l) = \frac{\prod_{r=1}^{B_l} n_r^{(l)}!}{B_{l-1}!} \binom{B_{l-1}-1}{B_l-1}^{-1} \frac{1}{B_{l-1}} \tag{15}$$

where $n_r^{(l)}$ denotes the size of group r at level l, with the boundary condition $B_0 = N$, representing the total number of nodes in the network.

The full prior over the hierarchical partition structure is obtained as

$$P(\{\boldsymbol{b}_l\}) = \prod_{l=1}^{L} P(\boldsymbol{b}_l) \tag{16}$$

This prior favors parsimonious hierarchical representations by penalizing excessive fragmentation (large numbers of groups), while remaining noninformative.

2.2.6 Full Joint Posterior for the Structural Prior

To construct the full unconditioned generative model, we compute the joint distribution of the network $\boldsymbol{A}$ and the entire nested hierarchy of latent parameters $\boldsymbol{\theta} = \{\boldsymbol{k}, \{\boldsymbol{e}_l\}, \{\boldsymbol{b}_l\}\}$.

The final formula with all the priors for the joint distribution of microcanonical HSBM is:

$$\begin{aligned} P(\boldsymbol{A}, \boldsymbol{k}, \{\boldsymbol{e}_l\}, \{\boldsymbol{b}_l\}) = P(\boldsymbol{A}|\boldsymbol{k}, \boldsymbol{e}_1, \boldsymbol{b}_1) \times P(\boldsymbol{k}|\boldsymbol{e}_1, \boldsymbol{b}_1) \\ \times P(\{\boldsymbol{e}_l\}|\{\boldsymbol{b}_l\}) \times P(\{\boldsymbol{b}_l\}). \end{aligned} \tag{17}$$

As we saw in Section 2.2.2, a crucial property of the microcanonical ensemble is that the parameters $\boldsymbol{k}$ and $\boldsymbol{e}$ are strictly deterministic functions of the network topology $\boldsymbol{A}$ and the block partition $\boldsymbol{b}$. Once a network and a group assignment are proposed, the degree sequence $\boldsymbol{k}$ and the block-to-block edge counts $\boldsymbol{e}$ are fixed mathematically. Consequently, the posterior depends only on $\boldsymbol{A}$ and $\boldsymbol{b}$, and the full structural joint probability $P(\boldsymbol{A}, \boldsymbol{\theta}) = P(\boldsymbol{A} \mid \boldsymbol{\theta})P(\boldsymbol{\theta})$ can be written solely is terms of the network and the block partition as $P(\boldsymbol{A} \mid \boldsymbol{b})P(\boldsymbol{b})$.

2.3 Description length and MCMC inference

To infer the optimal true network $\boldsymbol{A}$ and hierarchical block structure $\{\boldsymbol{b}_l\}$, we must maximize their joint posterior probability. Computationally, this is equivalent to minimizing its negative logarithm, which represents the Description Length (DL) of the network. In information theory, the DL represents the total number of bits required to transmit the network data. Following the framework of Peixoto (2014, 2017), this description length (Σ) can be cleanly partitioned into competing information-theoretic components:

$$\Sigma = \mathcal{M} + \mathcal{S} + \mathcal{L} \tag{18}$$

where:

- $\mathcal{M}$ (Measurement Error): The amount of information required to describe the discrepancies between the observations and the hypothesized true network.
- $\mathcal{S}$ (Structural Entropy): The amount of information required to describe the interaction structure of the true network, assuming the hierarchical community structure is already known.
- $\mathcal{L}$ (Model Complexity): The amount of information required to describe the community structure parameters.

By expanding these three components into their constituent generative steps (derived in previous sections), the full unconditioned description length becomes:

$$\Sigma(\boldsymbol{A}, \{\boldsymbol{b}_l\}) = \underbrace{-\ln P(\boldsymbol{x} \mid \boldsymbol{n}, \boldsymbol{A})}_{\mathcal{M}} + \underbrace{-\ln P(\boldsymbol{A} \mid \boldsymbol{k}, \boldsymbol{e}_1, \boldsymbol{b}_1)}_{\mathcal{S}} \\ + \underbrace{-\ln P(\boldsymbol{k} \mid \boldsymbol{e}_1, \boldsymbol{b}_1) - \ln P(\{\boldsymbol{e}_l\} \mid \{\boldsymbol{b}_l\}) - \ln P(\{\boldsymbol{b}_l\})}_{\mathcal{L}}. \tag{19}$$

Because the exact analytical calculation of the marginal probability across all possible networks and partitions is intractable, this landscape is explored using Markov Chain Monte Carlo (MCMC) sampling (Peixoto, 2018, 2020). When the algorithm proposes a local modification, the total change in the description length ($\Delta\Sigma$) dictates the acceptance probability via the Metropolis-Hastings criterion. A move is generally accepted if it yields a negative change in the description length ($\Delta\Sigma < 0$). This ensures the algorithm adheres to a parsimonious solution where it will only accept an increase in model complexity ($\mathcal{L}$) if it yields a substantially larger reduction in structural entropy ($\mathcal{S}$) or measurement error ($\mathcal{M}$).

In this framework, the microcanonical parameters (node degrees $\boldsymbol{k}$ and block-to-block edges $\boldsymbol{e}$) are deterministic counters that immediately reflect the current state of $\boldsymbol{A}$ and $\boldsymbol{b}$. The algorithm employs an alternating sampling scheme (Peixoto, 2018), decoupling the inference into two main phases:

- Updating the true network ($\boldsymbol{A}$): In this phase, the hierarchical community structure ($\boldsymbol{b}$) is fixed. The algorithm selects a random node pair (i, j) and proposes flipping its interaction state ($A_{ij} \to 1 - A_{ij}$). If an unobserved edge ($D_{ij} = 0$) is proposed as a true edge ($A_{ij} \to 1$), the degrees of nodes i and j increment, and the edge count between their respective blocks increments. The algorithm calculates $\Delta\Sigma$. The flip is accepted only if the structural SBM penalty of adding that edge to the topology is mathematically outweighed by the measurement model's combinatorial evidence for a false negative. Because the global error rates were integrated out analytically, this evidence evaluates natively via the global sums ($\mathcal{M}, \mathcal{X}, \mathcal{E}, \mathcal{T}$).
- Updating the community structure ($\boldsymbol{b}$): In this phase, the true network ($\boldsymbol{A}$) is fixed. The algorithm proposes modifying the block partitions using an efficient merge-split algorithm (Peixoto, 2020). When a node is moved to a new block, the degree sequence $\boldsymbol{k}$ remains identical, but the block edge counts $\boldsymbol{e}$ are recalculated. The algorithm frequently proposes placing a node into a completely empty block, or merging two existing blocks. If two blocks share highly similar interaction patterns, merging them leaves the topological entropy largely unchanged while significantly dropping the model complexity penalty ($-\ln P(\boldsymbol{b})$). The algorithm accepts the merge, naturally compressing the number of groups.

By jointly exploring the space of the network $\boldsymbol{A}$ and the hierarchical partitions $\boldsymbol{b}$, the HSBM disentangles sampling noise from community architecture, yielding a parsimonious and statistically principled reconstruction of the network.

It is important to emphasize that, under either a Bayesian or DL interpretation, multiple different network structures or partitions may yield nearly identical description lengths. Because of this, inference should not rely on a single solution. Instead, we should explore the solution landscape and report and/or average over an ensemble of highly probable configurations (Calatayud et al., 2019).

3 Beyond assortative communities

A persistent interpretation in network science is thinking of community detection as the same as assortative clustering. Assortative communities are groups of nodes that connect densely with each other and sparsely with the rest of the network. While some algorithms, such as modularity maximization, strictly search for assortative structures, the Stochastic Block Model (SBM) is agnostic to the interaction topology.

The topological flexibility of the SBM is entirely encapsulated within the block-to-block interaction matrix (represented by the expected edges counts $\boldsymbol{\omega}$ in the canonical formulation or the exact edge counts $\boldsymbol{e}$ in the microcanonical formulation). There is no constraint forcing the diagonal entries of this matrix (e_{rr}, representing within-block interactions) to be strictly greater than the off-diagonal entries (e_{rs}, representing between-block interactions). Because the MCMC algorithm seeks only to minimize the description length of the data, it will infer whatever configuration of $\boldsymbol{e}$ most efficiently compresses the inferred true network $\boldsymbol{A}$.

By allowing the block-to-block edge counts to freely vary, the degree-corrected SBM naturally identifies the major topological archetypes found in ecological networks:

- **Assortative (Modular) Structure**
  Mathematical Signature: Heavy diagonals ($e_{rr} \gg e_{rs}$).
  Ecological Meaning: This represents classic groups. In food webs, this might represent distinct spatial niches (e.g., benthic vs. pelagic species). In host-parasite networks, it often reflects co-evolutionary clusters where parasites strictly specialize on phylogenetically closely related hosts.

- **Disassortative (Multipartite/Bipartite) Structure**
  Mathematical Signature: Empty diagonals, heavy off-diagonals ($e_{rr} \approx 0,\ e_{rs} > 0$)
  Ecological Meaning: This defines bipartite ecological networks. In a plant-pollinator network, let r be the indicator for the plants' group and s the indicator for the bees' group. Because, plants do not pollinate plants ($e_{rr} = 0$), and bees do not pollinate bees ($e_{ss} = 0$), the interactions strictly occur between the groups ($e_{rs} > 0$). The SBM natively identifies this structure without requiring to explicitly flag the data as bipartite.

- **Core-Periphery (Nested) Structure**
  Mathematical Signature: Asymmetric interaction densities (e_{rr} is high, e_{rs} is moderate, and $e_{ss} \approx 0$)
  Ecological Meaning: This is typical of nestedness in mutualistic networks. Block r represents the "core" generalists (abundant plants and generalist pollinators that interact heavily with each other). Block s represents "peripheral" specialists. Specialists do not interact with other specialists ($e_{ss} \approx 0$), they interact with the generalists ($e_{rs} > 0$).

4 The influence of sampling effort

To understand how the total number of measurements or trials ($\boldsymbol{n}$ (introduced in Section 2.1) constrains the network reconstruction, we must examine the statistical relationship between the measurement data and the structural prior at the level of the node pairs (i, j):

- Total Ignorance ($n_{ij} = 0$): The measurement model has no statistical strength. In this case, empirical ignorance is total, and the framework engages in pure structural imputation. It predicts the probability of the unmeasured interaction by "borrowing strength" from the macroscopic HSBM structure.

- Binary/Single Trial ($n_{ij} = 1$): If $x_{ij} = 0$, the measurement model weakly suggests the edge doesn't exist, but it is highly uncertain (it could easily be a false negative). The HSBM prior heavily influences the final prediction. If the inferred hierarchical structure strongly expects an edge between those specific nodes, it will override the single $x_{ij} = 0$ observation and flag it as a missing link. This case is the default assumption in *sabinaHSBM* package and was detailed in the Section 2.1.1.

- High sampling effort ($n_{ij} \gg 1$): If a pair is heavily sampled (e.g. $n_{ij} = 50$) and yields zero observed interactions ($x_{ij} = 0$) the measurement model provides strong evidence of a true absence. The description length for claiming 50 consecutive false negatives is high. Consequently, the measurement data will overpower the HSBM prior, driving the final edge probability to near zero, even if the nodes belong to compatible interaction groups.

By default, the *sabinaHSBM* package assumes a single-trial case ($n_{ij} = 1$ and $x_{ij} = D_{ij}$ for all pairs). However, it allows users to explicitly provide empirical sampling data by supplying a data frame of observed interactions (x_{ij}) and total trials (n_{ij}) to the `custom_nx` argument in the `hsbm.input()` function. The package safely manages these custom inputs during cross-validation temporarily forcing held-out links to $n = 1$ and $x = 0$. This guarantees that user-defined sampling efforts do not overwrite the cross-validation folds.

The use of `custom_nx`, allows for fine-grained control over heterogeneous sampling efforts. However, we caution researchers to treat the sampling effort variables transparently and ensure that these parameters are entirely justified by the data at hand. Altering these values arbitrarily or based on weak justification will bias the inference process and potentially lead to erroneous conclusions.

4.1 The distinction between observed absences and missing data

In ecology, there is a persistent confusion between entries coded as 0 (an observed absence) and entries coded as NA (unmeasured or missing data). Because standard adjacency matrices often conflate these two states by coding both as 0, critical structural uncertainty is frequently misrepresented. So we extend the information given in the previous section to this specific scenario.

- Missing Data ($NA \to (n_{ij} = 0, x_{ij} = 0)$): If a species pair was never sampled (e.g., due to disjoint spatial or temporal sampling ranges), it represents pure missing data (NA). By setting $n_{ij} = 0$, the model is informed of total empirical ignorance. It will not penalize the global false negative rate, but will instead impute the probability of the link purely by "borrowing strength" from the macroscopic HSBM structure.
- Observed Absence ($0 \to (n_{ij} > 0, x_{ij} = 0)$): If a species pair was actively monitored but no interaction was ever recorded, it is a true absence. Users should set $n_{ij} > 0$ and $x_{ij} = 0$. This is especially relevant for cases when $n_{ij} \gg 1$ since it acts as strong empirical evidence against the link, demanding substantial structural counter-evidence before the model will predict a missing edge.

4.2 Asymmetry of detection: note on rare observations

A common concern is whether the measurement model will penalize rare, hard-to-detect interactions ($x_{ij} = 1, n_{ij} \gg 1$). In particular, if an interaction is observed only once out of ten independent trials ($n_{ij} = 10, x_{ij} = 1$), one might assume that the empirical evidence for a true edge is weak. However, the Beta-Binomial measurement model naturally protects these rare observations due to the severe asymmetry between error rates. Because interaction matrices are generally sparse, the inferred global false positive rate (q) is mathematically near zero, whereas the false negative rate (p) is typically much larger. Consequently, the likelihood of missing a true interaction 9 times out of 10 is statistically plausible, while the likelihood of falsely recording a spurious interaction even once is extremely low. Thus, the measurement model inherently respects the empirical presence of a rare observation. It demands overwhelming counter-evidence from the HSBM structural prior before it will override the data and classify any observed interaction as spurious. This can be illustrated by the cost of finding spurious for single-trial networks, which we do in the next section.

5 The cost of finding spurious and missing links in sparse networks

Because networks are generally sparse, the measurement model mathematically constrains the probability of false positives to highly unlikely.

Assuming a single trial scenario ($n_{ij} = 1,\ x_{ij} \in \{0, 1\}$), and a non-informative Beta(1, 1) prior for the error rates, the model estimates the false positive rate q by evaluating the proposed true network $\boldsymbol{A}$. Specifically, it looks at the ratio of False Positives (FP: pairs where $A_{ij} = 0$ but $x_{ij} = 1$) to True Negatives (TN: pairs where $A_{ij} = 0$ and $x_{ij} = 0$). Because the Beta prior and Binomial likelihood are mathematically conjugate, the posterior distribution for the false positive rate q conditioned on the observed network data $\boldsymbol{x}$ and true network $\boldsymbol{A}$ is a new Beta distribution:

$$\begin{aligned} P(q|\boldsymbol{x}, \boldsymbol{A}) &= \mathrm{Beta}(FP + \alpha_{\mathrm{prior}},\, TN + \beta_{\mathrm{prior}}) \\ &= \mathrm{Beta}(FP + 1,\, TN + 1) \end{aligned} \tag{20}$$

The expected value of any Beta(α, β) distribution is exactly $\frac{\alpha}{\alpha+\beta}$. Plugging in our posterior parameters, the expected value of the false positive rate becomes:

$$\mathbb{E}[q] = \frac{FP + 1}{(FP + 1) + (TN + 1)} = \frac{FP + 1}{FP + TN + 2} \tag{21}$$

To illustrate this, consider an undirected ecological network of 100 species with no intraspecific interactions, yielding $\binom{100}{2} = 4,950$ unique pairs. Assume we observe 200 interactions (1s) and 4,750 non-interactions (0s). If the MCMC algorithm proposes flagging a single observed edge as a spurious link ($FP = 1$), it evaluates this against the 4,750 non-edges (TN). The expected value is:

$$\mathbb{E}[q] = \frac{1 + 1}{1 + 4750 + 2} = \frac{2}{4753} \approx 0.00042 \tag{22}$$

Because TN is massively large in sparse networks, the "+1" and "+2" from the uniform prior become mathematically negligible. The expected value converges to the simple heuristic $\frac{FP}{FP+TN}$. More importantly, because the sample size for non-edges is so large, the variance of this Beta distribution shrinks to near zero. Since the false positive rate is near 0.04%, it is statistically highly unlikely for any of the 200 observed interactions to be spurious (i.e. erroneously recorded). So, to delete an observed edge, the model must pay a huge statistical penalty in the measurement likelihood.

In addition, the HSBM tends to reinforce the existence of observed edges during inference. When an observed edge ($x_{ij} = 1$) is fed into the model, the HSBM can place species i and j into compatible blocks specifically because they share that edge. Because they are now in compatible blocks, the HSBM structural prior increases the posterior support for the edge. Therefore, for an observed edge to be flagged as spurious, its presence must be very anti-structural, violating established macroscopic community rules so strongly that the topological penalty of keeping it outweighs the massive measurement penalty of deleting it.

Conversely, adding a missing link ($0 \to 1$) is statistically "cheap." Because there are thousands of observed zeros in the matrix, the model can easily classify a few of them as False Negatives (*FN*) without drastically inflating the overall false negative rate p. Furthermore, proposing a missing link between two already highly connected blocks directly improves the HSBM's internal structural entropy, efficiently lowering the total description length.

Ultimately, the framework is mathematically highly conservative regarding edge deletion, but remains flexible and adaptive regarding missing link imputation.

If spurious links are suspected in a network and the HSBM framework fails to flag any spurious link, the *sabinaHSBM* offers the possibility to rank observed links by their probabilities in ascending order using the `top_links` function with the argument `edge_type = "documented"`. The first links in the rank are the observed links that had a lower link probability, so these can be targeted for reliability assessment. For a practical demonstration, check the Supporting Information S3 included in the supplementary material of the manuscript.

6 Technical details of the `hsbm.predict()` function

6.1 MCMC sampling process

The inference procedure in the *sabinaHSBM* package relies on the `MeasuredBlockModel` implementation in the *graph-tool* Python module. The MCMC process inside the package unfolds in two stages: an equilibration phase and a sampling phase.

During the equilibration phase, the model iteratively searches for convergence. In each iteration, the algorithm performs 10 sweeps of proposals to move nodes between blocks and flip edges, accepting or rejecting these moves based on reductions in the Description Length (DL). Convergence is assessed using the `wait` argument (default: 1,000 iterations), which defines the minimum number of successive iterations during which the minimum and maximum values of the DL must remain stable. This period of stability must occur twice to ensure the chain has not simply stalled in a local fluctuation.

Once equilibration is achieved, the posterior sampling phase begins. The model performs a user-defined number of additional iterations, controlled by the `iter` argument (default: 10,000). To mitigate autocorrelation, only the final network configuration ($\boldsymbol{A}$) and partition ($\boldsymbol{b}$) at the end of each iteration are retained. These retained samples form the empirical posterior distribution, which is used to calculate the final link probabilities.

6.2 Link prediction methods

Recall that while the measurement model evaluates quantitative sampling effort (x_{ij} observed interaction events out of n_{ij} trials), standard network reconstruction defaults to operating on a binary empirical adjacency matrix $\boldsymbol{D}$ as its input data. In this binary context, $D_{ij} = 1$ simply indicates that an interaction was observed at least once, and $D_{ij} = 0$ indicates it was never observed. By default, the *sabinaHSBM* package translates this binary input into a single-trial assumption ($x_{ij} = D_{ij}$ and $n_{ij} = 1$). However, users can supply different sampling effort count data for any pair (where $n_{ij} \geq 0$ and $0 \leq x_{ij} \leq n_{ij}$) using the `custom_nx` argument.

Two methods for link prediction are available via the *method* argument:

The `"conditional_missing"` method (conditional probability averaging) is a classical gap-filling approach that focuses exclusively on scoring unobserved links (i.e., those with $D_{ij} = 0$). Specifically, for each unobserved node pair (i, j), it computes the average conditional probability that a link exists, given the inferred block structure. After the initial MCMC equilibration phase (controlled with `wait` argument), a sequence of N posterior block partitions — indexed as $\boldsymbol{b}^{(k)}$ for $k = 1, ..., N$ — is obtained during the sampling phase (controlled with `iter` argument). For each partition k, the package calls the `MeasuredBlockState.get_edges_prob()` function in *graph-tool* to compute the conditional probability $P(A_{ij} = 1|\boldsymbol{b}^{(k)})$. It then averages these values across all N samples to yield the final link probability:

$$p_{ij} = \frac{1}{N} \sum_{k=1}^{N} P(A_{ij} = 1|\boldsymbol{b}^{(k)}) \tag{23}$$

By relying exclusively on the inferred block structure, the `"conditional_missing"` method assigns each unobserved link a score based on its likelihood under the generative model. It retains the classical HSBM link-prediction paradigm (Ghasemian et al., 2020), allowing direct comparison with other existing algorithms. However, this method does not yield true marginal probabilities, and evaluating the analytical probability for every non-edge is substantially more computationally intensive than the `"marginal_all"` method (see Section 7). For these reasons, *sabinaHSBM* restricts this method to estimating probabilities for unobserved links ($D_{ij} = 0$) only.

The `"marginal_all"` method (marginal posterior probability estimation) is a full network reconstruction approach that computes the marginal posterior probability $P(A_{ij} = 1|\boldsymbol{D})$ for observed ($D_{ij} = 1$) and unobserved ($D_{ij} = 0$) links. After the MCMC equilibration phase (controlled by the `wait` argument), we draw a sequence of N samples of the latent true network $\boldsymbol{A}$ — indexed as $\boldsymbol{A}^{(k)}$ for $k = 1, ..., N$ — during the sampling phase (set via *iter* argument). For each sample k we use the `MeasuredBlockState.collect_marginal()` function from *graph-tool* to extract the latent network $\boldsymbol{A}^{(k)}$ and record every link presence $A_{ij}^{(k)} = 1$. These link presences are averaged to obtain:

$$p_{ij} = \frac{1}{N} \sum_{k=1}^{N} A_{ij}^{(k)} \tag{24}$$

where $A_{ij}^{(k)} = 1$ if the link (i, j) exists in the k-th sampled latent network, and 0 otherwise. The resulting p_{ij} directly reflects the accumulated evidence for the existence of the link (i, j) across the joint posterior. The `"marginal_all"` method delivers a marginal probability matrix that simultaneously uncovers missing links (false negatives) and spurious links (false positives), making it the method of choice for network diagnostics and quality assessment.

Because this method uses sampled networks rather than focusing on individual link probabilities conditioned on an inferred block structure, highly unlikely links may never appear in any sample $\boldsymbol{A}^{(k)}$, yielding $p_{ij} \approx 0$. The *sabinaHSBM* package lets you handle these cases flexibly via the `na_treatment` argument on the `hsbm.reconstructed()` function, ensuring no uncertainty is lost or misrepresented in downstream analyses.

6.2.1 Computational cost of both methods

The substantial difference in computational cost between the two prediction methods arises from how they evaluate the unobserved edges of the network. The `"conditional_missing"` method must iterate over every single unobserved pair ($D_{ij} = 0$) to compute the conditional probability of an edge existing. Because it must evaluate this for the all non-edges, the computational cost scales quadratically with the number of nodes, $\mathcal{O}(V^2)$. In contrast, the `"marginal_all"` method can leverage the common sparsity of the generated networks. Rather than evaluating analytical probabilities for individual non-edges, the algorithm sweeps the full discrete network $\boldsymbol{A}^{(k)}$ generated at each MCMC step and simply counts the occurrences of the edges ($A_{ij}^{(k)} = 1$). Because this approach completely ignores non-edges its computational cost scales with the network density, $\mathcal{O}(E)$. Benchmarks for both methods can be seen in section 7.

6.3 Key function parameters

The `hsbm.predict()` process is controlled by several arguments:

- `wait`: The number of successive, stable MCMC iterations required to assess convergence during the equilibration phase (default: 1,000).
- `iter`: The number of posterior samples generated after the equilibration phase (default: 10,000).

- `n_cores`: Enables parallel computation across folds, to speed up execution on large datasets or multiple folds (default: 1).
- `rnd_seed`: An optional argument to fix the random seed for the HSBM inference to ensure reproducibility.
- `elist_i`: Selects a specific cross-validation fold for computation. By default, predictions are computed across all folds.

6.4 Additional function outputs

Optionally, the function allows saving the results and *graph-tool* objects as Python pickle files with `save_pickle = TRUE`. These files are stored in the working directory under names like *hsbm_res_foldi.pkl*, where i denotes the fold index. Also, a simple hierarchical edge bundling figure for each fold can be saved in the working directory, using the argument `save_plots = TRUE`.

7 Benchmarks

We generated 11 parameterizations of bipartite networks spanning a gradient of connectance using a probabilistic, modular network model. An exponential phylogenetic decay function reduced the probability of interactions between evolutionarily distant lineages, generating modules of closely related species. A global connectivity parameter allowed highly connected generalists to bridge modules, whereas an affinity-shape parameter determined the internal distribution of interactions within each module.

For each parameterization, we generated five replicate networks, yielding a total of 55 simulated networks. Each network was analyzed with *sabinaHSBM* using five-fold cross-validation to compare the computation times of the `"marginal_all"` and `"conditional_missing"` reconstruction methods. The default threshold (`"roc_youden"`) was used, and for the `"marginal_all"` method the argument `rm_documented = FALSE` was specified in the `hsbm.reconstructed` function.

The code was run in a Dell Inspiron notebook with Intel Core i7- 7500U CPU. Benchmarks are for the entire prediction and reconstruction workflow using a single core.

A plot of the interaction network for one network made with the parameterization with the highest number of links (par11) is shown below.

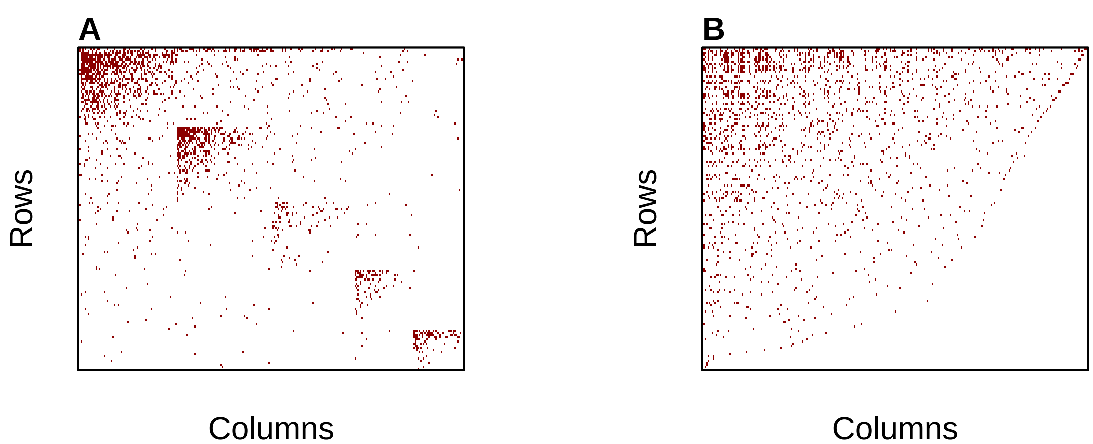

Fig. S1. Example of a simulated bipartite interaction network generated under the benchmark framework. The network shown corresponds to the parameterization with the highest number of interactions (par11). Rows and columns represent the two node sets of the bipartite network, and filled cells indicate observed interactions. (A) Illustrates the structure of the network highlighting the modules used for network generation. (B) Is the same network as (A) but ordered by row frequencies.

The benchmark results show that the "`conditional_missing`" method scales linearly with the number of interactions, requiring nearly 300 minutes to process networks with ~2,000 links. The "`marginal_all`" method exhibits a sigmoidal curve, maximizing execution time at roughly 70 minutes for the same network size, offering a more than four-fold reduction in total runtime (Fig. S2A, S2B). Peak memory consumption for "`conditional_missing`" also increases linearly up to nearly 5 GB of RAM at 2,000 links (Fig. S2C). In contrast, `marginal_all` shows an efficient memory profile, using around ~1.4 GB of RAM regardless of network size (Fig. S2D).

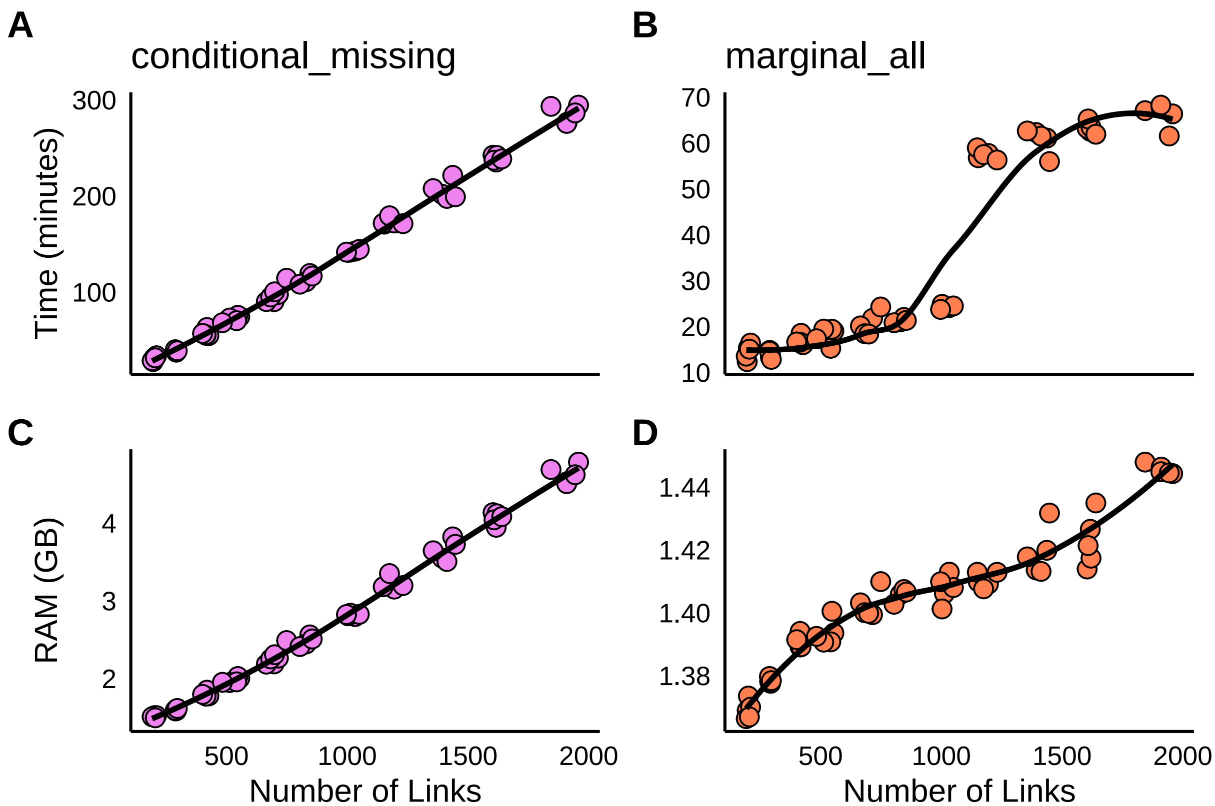

Fig. S2. Computational performance of the two network reconstruction methods implemented in sabinaHSBM across simulated networks of increasing size. (A) Total computation time for the 'conditional_missing' method. (B) Total computation time for the 'marginal_all' method. (C) Peak memory (RAM) consumption as a function of the number of links for the 'conditional_missing' method. (D) Peak memory (RAM) consumption as a function of the number of links for the 'marginal_all' method.

While "`conditional_missing`" achieves a higher recovery of held-out interactions it generates a significantly larger number of total predicted links (Table S7; Fig. S3A). The "`marginal_all`" method provides a conservative reconstruction, recovering ~20% of held-out link while constraining false positive inflation (Table S7).

Table S7. Summary statistics for the benchmark simulations and reconstruction performance. For each reconstruction method ('conditional_missing' and 'marginal_all') and network parameterization (par01 to par11), the table reports in this order, the mean number of links , connectance, modularity, nestedness, cross-validation AUC, proportion of held-out interactions recovered (pred_held_ones), and total number of predicted interactions (pred_ones). Values are averaged across the five replicate networks generated for each parameterization.

method	**par**	**links**	**connectance**	**mod**	**nest**	**auc**	**pred_held_ones**	**pred_ones**
conditional_missing	par01	200.8	0.08	0.50	16.72	0.89	0.91	482.4
conditional_missing	par02	291.6	0.07	0.48	15.49	0.90	0.88	670.6
conditional_missing	par03	415.4	0.06	0.44	14.84	0.91	0.91	1356.8
conditional_missing	par04	528.0	0.06	0.45	14.75	0.91	0.88	1615.0
conditional_missing	par05	691.8	0.06	0.45	14.23	0.90	0.85	2150.8
conditional_missing	par06	816.6	0.05	0.46	13.27	0.91	0.86	2678.8
conditional_missing	par07	1019.2	0.05	0.44	12.71	0.90	0.84	3454.4
conditional_missing	par08	1180.6	0.05	0.45	12.19	0.90	0.81	3743.0
conditional_missing	par09	1409.4	0.05	0.45	12.15	0.91	0.85	5712.4
conditional_missing	par10	1617.8	0.05	0.44	12.33	0.91	0.86	6700.6
conditional_missing	par11	1913.2	0.04	0.43	11.89	0.90	0.81	7102.2
marginal_all	par01	200.8	0.08	0.50	16.72	1.00	0.19	207.2
marginal_all	par02	291.6	0.07	0.47	15.49	1.00	0.22	294.6
marginal_all	par03	415.4	0.06	0.43	14.84	1.00	0.20	431.0
marginal_all	par04	528.0	0.06	0.45	14.75	1.00	0.21	549.2
marginal_all	par05	691.8	0.06	0.45	14.23	1.00	0.19	717.8
marginal_all	par06	816.6	0.05	0.45	13.27	1.00	0.14	830.8
marginal_all	par07	1019.2	0.05	0.44	12.71	1.00	0.15	1045.2
marginal_all	par08	1180.6	0.05	0.45	12.19	1.00	0.15	1218.6
marginal_all	par09	1409.4	0.05	0.45	12.15	1.00	0.14	1461.4
marginal_all	par10	1617.8	0.05	0.44	12.33	1.00	0.17	1703.6
marginal_all	par11	1913.2	0.04	0.43	11.89	1.00	0.20	2046.2

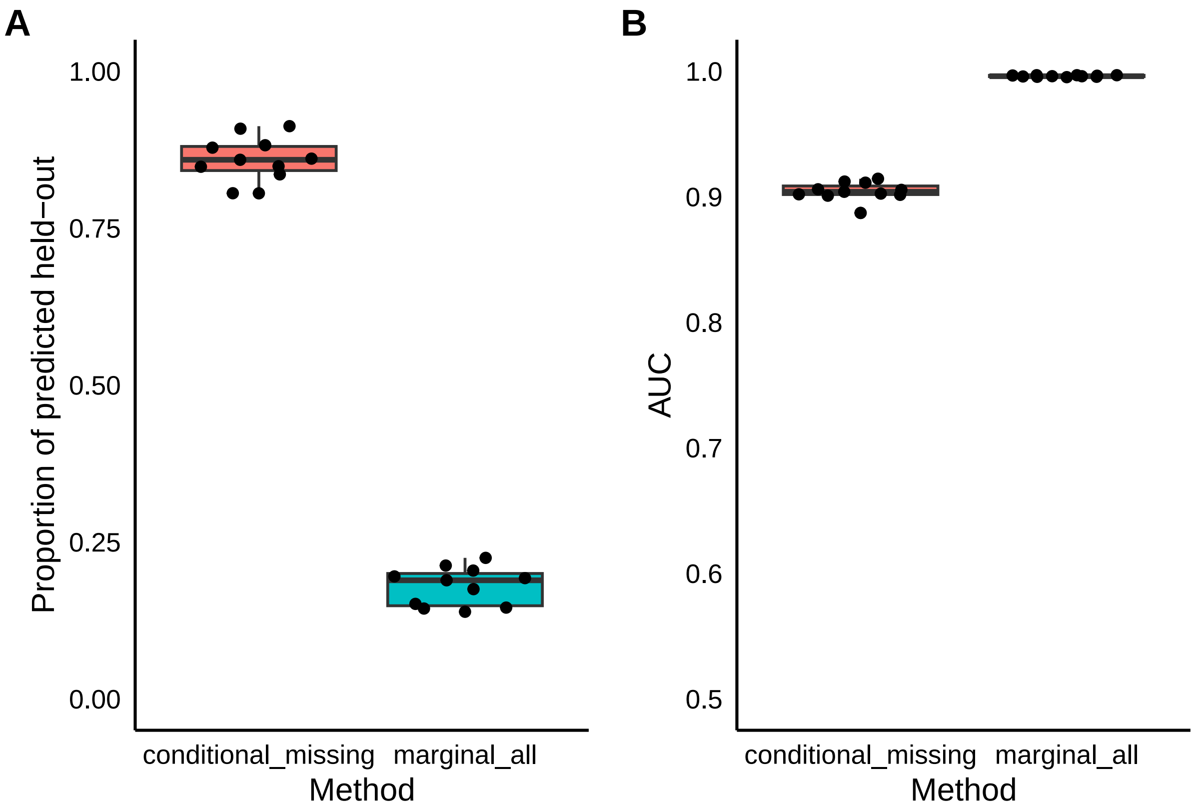

Fig. S3. Predictive performance of the two reconstruction methods evaluated using five-fold cross-validation across the 55 simulated networks. The default Youden threshold ('roc_youden') was used. (A) Proportion of held-out interactions correctly recovered by each method. (B) Area under the receiver operating characteristic curve (AUC). The points are average values for each parameterization.

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